

BEE 4750 Lab 1: Julia and GitHub Basics

Name:

ID:

Due Date

Thursday, 8/28/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

```
Activating project at `c:\Users\Donny\lab1-dc2266`
  Updating registry at `C:\Users\Donny\.julia\registries\General.toml`
Installed x265_jll          v4.1.0+0
Installed libfdk_aac_jll    v2.0.4+0
Installed LERC_jll          v4.0.1+0
Installed GR_jll            v0.73.17+0
Installed Libmount_jll      v2.41.1+0
Installed Preferences        v1.5.0
Installed JpegTurbo_jll     v3.1.2+0
Installed libdecor_jll      v0.2.2+0
Installed Opus_jll          v1.5.2+0
Installed LoggingExtras      v1.1.0
Installed Unitful            v1.24.0
Installed Xorg_xkbcomp_jll  v1.4.7+0
```

Installed RelocatableFolders	v1.0.1
Installed Contour	v0.6.3
Installed Xorg_xcb_util_image_jll	v0.4.1+0
Installed Measures	v0.3.2
Installed Grisu	v1.0.2
Installed ConcurrentUtilities	v2.5.0
Installed Xorg_xcb_util_wm_jll	v0.4.2+0
Installed PlotUtils	v1.4.3
Installed DelimitedFiles	v1.9.1
Installed OpenSSL	v1.5.0
Installed libsodium_jll	v1.0.21+0
Installed HTTP	v1.10.17
Installed RecipesPipeline	v0.6.12
Installed Xorg_libSM_jll	v1.2.6+0
Installed Xorg_xcb_util_jll	v0.4.1+0
Installed Cairo_jll	v1.18.5+0
Installed Xorg_libxkbfile_jll	v1.1.3+0
Installed Fontconfig_jll	v2.17.1+0
Installed Statistics	v1.11.1
Installed Xorg_libXinerama_jll	v1.1.6+0
Installed Xorg_libXau_jll	v1.0.13+0
Installed EpollShim_jll	v0.0.20230411+1
Installed ColorSchemes	v3.30.0
Installed IrrationalConstants	v0.2.4
Installed Missings	v1.2.0
Installed FFMPEG	v0.4.4
Installed Showoff	v1.0.3
Installed GR	v0.73.17
Installed JSON	v0.21.4
Installed Pango_jll	v1.56.3+0
Installed PtrArrays	v1.3.0
Installed SimpleBufferStream	v1.2.0
Installed Xorg_xcb_util_keysyms_jll	v0.4.1+0
Installed xkbcommon_jll	v1.9.2+0
Installed Bzip2_jll	v1.0.9+0
Installed HarfBuzz_jll	v8.5.1+0
Installed XZ_jll	v5.8.1+0
Installed PlotThemes	v3.3.0
Installed IJulia	v1.30.0
Installed NaNMath	v1.1.3
Installed fzf_jll	v0.61.1+0
Installed LZ0_jll	v2.10.3+0
Installed ZeroMQ_jll	v4.3.6+0

Installed TranscodingStreams	v0.11.3
Installed GLFW_jll	v3.4.0+2
Installed FriBidi_jll	v1.0.17+0
Installed UnicodeFun	v0.4.1
Installed MbedTLS	v1.1.9
Installed x264_jll	v10164.0.1+0
Installed FreeType2_jll	v2.13.4+0
Installed StatsAPI	v1.7.1
Installed StatsBase	v0.34.6
Installed JLFzf	v0.1.11
Installed CodecZlib	v0.7.8
Installed DataStructures	v0.19.1
Installed Colors	v0.13.1
Installed Dbus_jll	v1.16.2+0
Installed libpng_jll	v1.6.50+0
Installed mtdev_jll	v1.1.7+0
Installed Xorg_libxcb_jll	v1.17.1+0
Installed ExceptionUnwrapping	v0.1.11
Installed ColorTypes	v0.12.1
Installed Scratch	v1.3.0
Installed libaom_jll	v3.12.1+0
Installed eudev_jll	v3.2.14+0
Installed Xorg_libXext_jll	v1.3.7+0
Installed Conda	v1.10.2
Installed Zstd_jll	v1.5.7+1
Installed TensorCore	v0.1.1
Installed ZMQ	v1.4.1
Installed Xorg_xcb_util_cursor_jll	v0.1.5+0
Installed Plots	v1.40.19
Installed Libtiff_jll	v4.7.1+0
Installed Format	v1.3.7
Installed Expat_jll	v2.7.1+0
Installed Parsers	v2.8.3
Installed Libffi_jll	v3.4.7+0
Installed JLLWrappers	v1.7.1
Installed ColorVectorSpace	v0.11.0
Installed OrderedCollections	v1.8.1
Installed libinput_jll	v1.28.1+0
Installed libevdev_jll	v1.13.4+0
Installed Xorg_libXi_jll	v1.8.3+0
Installed Vulkan_Loader_jll	v1.3.243+0
Installed Xorg_libXrender_jll	v0.9.12+0
Installed Ogg_jll	v1.3.6+0

Installed Qt6ShaderTools_jll	v6.8.2+1
Installed Reexport	v1.2.2
Installed Qt6Declarative_jll	v6.8.2+1
Installed Xorg_libXcursor_jll	v1.2.4+0
Installed Libuuid_jll	v2.41.1+0
Installed Xorg_xcb_util_renderutil_jll	v0.3.10+0
Installed AliasTables	v1.1.3
Installed LogExpFunctions	v0.3.29
Installed MacroTools	v0.5.16
Installed DocStringExtensions	v0.9.5
Installed Xorg_xtrans_jll	v1.6.0+0
Installed Xorg_libICE_jll	v1.1.2+0
Installed Pixman_jll	v0.44.2+0
Installed Wayland_jll	v1.24.0+0
Installed libass_jll	v0.17.4+0
Installed Graphite2_jll	v1.3.15+0
Installed Latexify	v0.16.9
Installed StableRNGs	v1.0.3
Installed FFMPEG_jll	v7.1.1+0
Installed BitFlags	v0.1.9
Installed LLVMOpenMP_jll	v18.1.8+0
Installed Xorg_xkeyboard_config_jll	v2.44.0+0
Installed Xorg_libXrandr_jll	v1.5.5+0
Installed OpenSSL_jll	v3.5.2+0
Installed FixedPointNumbers	v0.8.5
Installed DataAPI	v1.16.0
Installed Xorg_libXfixes_jll	v6.0.1+0
Installed Libiconv_jll	v1.18.0+0
Installed LAME_jll	v3.100.3+0
Installed RecipesBase	v1.3.4
Installed Qt6Base_jll	v6.8.2+1
Installed GettextRuntime_jll	v0.22.4+0
Installed Qt6Wayland_jll	v6.8.2+1
Installed PrecompileTools	v1.2.1
Installed Glib_jll	v2.84.3+0
Installed libvorbis_jll	v1.3.8+0
Installed URIs	v1.6.1
Installed VersionParsing	v1.3.0
Installed LaTeXStrings	v1.4.0
Installed Requires	v1.3.1
Installed Libglvnd_jll	v1.7.1+1
Installed Xorg_libX11_jll	v1.8.12+0
Installed Xorg_libXdmcp_jll	v1.1.6+0

```

Installed Unzip                v0.2.0
Installed SortingAlgorithms    v1.2.2
Installed UnitfulLatexify      v1.7.0
  Updating `C:\Users\Donny\lab1-dc2266\Project.toml`
[7073ff75] + IJulia v1.30.0
[b964fa9f] + LaTeXStrings v1.4.0
[91a5bcdd] + Plots v1.40.19
  Updating `C:\Users\Donny\lab1-dc2266\Manifest.toml`
[66dad0bd] + AliasTables v1.1.3
[d1d4a3ce] + BitFlags v0.1.9
[944b1d66] + CodecZlib v0.7.8
[35d6a980] + ColorSchemes v3.30.0
[3da002f7] + ColorTypes v0.12.1
[c3611d14] + ColorVectorSpace v0.11.0
[5ae59095] + Colors v0.13.1
[f0e56b4a] + ConcurrentUtilities v2.5.0
[8f4d0f93] + Conda v1.10.2
[d38c429a] + Contour v0.6.3
[9a962f9c] + DataAPI v1.16.0
[864edb3b] + DataStructures v0.19.1
[8bb1440f] + DelimitedFiles v1.9.1
[ffbed154] + DocStringExtensions v0.9.5
[460bff9d] + ExceptionUnwrapping v0.1.11
[c87230d0] + FFMPEG v0.4.4
[53c48c17] + FixedPointNumbers v0.8.5
[1fa38f19] + Format v1.3.7
[28b8d3ca] + GR v0.73.17
[42e2da0e] + Grisu v1.0.2
[cd3eb016] + HTTP v1.10.17
[7073ff75] + IJulia v1.30.0
[92d709cd] + IrrationalConstants v0.2.4
[1019f520] + JLFzf v0.1.11
[692b3bcd] + JLLWrappers v1.7.1
[682c06a0] + JSON v0.21.4
[b964fa9f] + LaTeXStrings v1.4.0
[23fbelc1] + Latexify v0.16.9
[2ab3a3ac] + LogExpFunctions v0.3.29
[e6f89c97] + LoggingExtras v1.1.0
[1914dd2f] + MacroTools v0.5.16
[739be429] + MbedTLS v1.1.9
[442fdcdd] + Measures v0.3.2
[e1d29d7a] + Missings v1.2.0
[77ba4419] + NaNMath v1.1.3

```

[4d8831e6] + OpenSSL v1.5.0
 [bac558e1] + OrderedCollections v1.8.1
 [69de0a69] + Parsers v2.8.3
 [ccf2f8ad] + PlotThemes v3.3.0
 [995b91a9] + PlotUtils v1.4.3
 [91a5bcdd] + Plots v1.40.19
 [aea7be01] + PrecompileTools v1.2.1
 [21216c6a] + Preferences v1.5.0
 [43287f4e] + PtrArrays v1.3.0
 [3cdcf5f2] + RecipesBase v1.3.4
 [01d81517] + RecipesPipeline v0.6.12
 [189a3867] + Reexport v1.2.2
 [05181044] + RelocatableFolders v1.0.1
 [ae029012] + Requires v1.3.1
 [6c6a2e73] + Scratch v1.3.0
 [992d4aef] + Showoff v1.0.3
 [777ac1f9] + SimpleBufferStream v1.2.0
 [a2af1166] + SortingAlgorithms v1.2.2
 [860ef19b] + StableRNGs v1.0.3
 [10745b16] + Statistics v1.11.1
 [82ae8749] + StatsAPI v1.7.1
 [2913bbd2] + StatsBase v0.34.6
 [62fd8b95] + TensorCore v0.1.1
 [3bb67fe8] + TranscodingStreams v0.11.3
 [5c2747f8] + URIs v1.6.1
 [1cfade01] + UnicodeFun v0.4.1
 [1986cc42] + Unitful v1.24.0
 [45397f5d] + UnitfulLatexify v1.7.0
 [41fe7b60] + Unzip v0.2.0
 [81def892] + VersionParsing v1.3.0
 [c2297ded] + ZMQ v1.4.1
 [6e34b625] + Bzip2_jll v1.0.9+0
 [83423d85] + Cairo_jll v1.18.5+0
 [ee1fde0b] + Dbus_jll v1.16.2+0
 [2702e6a9] + EpollShim_jll v0.0.20230411+1
 [2e619515] + Expat_jll v2.7.1+0
 [b22a6f82] + FFMPEG_jll v7.1.1+0
 [a3f928ae] + Fontconfig_jll v2.17.1+0
 [d7e528f0] + FreeType2_jll v2.13.4+0
 [559328eb] + FriBidi_jll v1.0.17+0
 [0656b61e] + GLFW_jll v3.4.0+2
 [d2c73de3] + GR_jll v0.73.17+0
 [b0724c58] + GettextRuntime_jll v0.22.4+0

[7746bdde] + Glib_jll v2.84.3+0
 [3b182d85] + Graphite2_jll v1.3.15+0
 [2e76f6c2] + HarfBuzz_jll v8.5.1+0
 [aacddb02] + JpegTurbo_jll v3.1.2+0
 [c1c5ebd0] + LAME_jll v3.100.3+0
 [88015f11] + LERC_jll v4.0.1+0
 [1d63c593] + LLVMOpenMP_jll v18.1.8+0
 [dd4b983a] + LZ0_jll v2.10.3+0
 [e9f186c6] + Libffi_jll v3.4.7+0
 [7e76a0d4] + Libglvnd_jll v1.7.1+1
 [94ce4f54] + Libiconv_jll v1.18.0+0
 [4b2f31a3] + Libmount_jll v2.41.1+0
 [89763e89] + Libtiff_jll v4.7.1+0
 [38a345b3] + Libuuid_jll v2.41.1+0
 [e7412a2a] + Ogg_jll v1.3.6+0
 [458c3c95] + OpenSSL_jll v3.5.2+0
 [91d4177d] + Opus_jll v1.5.2+0
 [36c8627f] + Pango_jll v1.56.3+0
 [30392449] + Pixman_jll v0.44.2+0
 [c0090381] + Qt6Base_jll v6.8.2+1
 [629bc702] + Qt6Declarative_jll v6.8.2+1
 [ce943373] + Qt6ShaderTools_jll v6.8.2+1
 [e99dba38] + Qt6Wayland_jll v6.8.2+1
 [a44049a8] + Vulkan_Loader_jll v1.3.243+0
 [a2964d1f] + Wayland_jll v1.24.0+0
 [ffd25f8a] + XZ_jll v5.8.1+0
 [f67eecfb] + Xorg_libICE_jll v1.1.2+0
 [c834827a] + Xorg_libSM_jll v1.2.6+0
 [4f6342f7] + Xorg_libX11_jll v1.8.12+0
 [0c0b7dd1] + Xorg_libXau_jll v1.0.13+0
 [935fb764] + Xorg_libXcursor_jll v1.2.4+0
 [a3789734] + Xorg_libXdmcp_jll v1.1.6+0
 [1082639a] + Xorg_libXext_jll v1.3.7+0
 [d091e8ba] + Xorg_libXfixes_jll v6.0.1+0
 [a51aa0fd] + Xorg_libXi_jll v1.8.3+0
 [d1454406] + Xorg_libXinerama_jll v1.1.6+0
 [ec84b674] + Xorg_libXrandr_jll v1.5.5+0
 [ea2f1a96] + Xorg_libXrender_jll v0.9.12+0
 [c7cfdc94] + Xorg_libxcb_jll v1.17.1+0
 [cc61e674] + Xorg_libxkbfile_jll v1.1.3+0
 [e920d4aa] + Xorg_xcb_util_cursor_jll v0.1.5+0
 [12413925] + Xorg_xcb_util_image_jll v0.4.1+0
 [2def613f] + Xorg_xcb_util_jll v0.4.1+0

[975044d2] + Xorg_xcb_util_keysyms_jll v0.4.1+0
[0d47668e] + Xorg_xcb_util_renderutil_jll v0.3.10+0
[c22f9ab0] + Xorg_xcb_util_wm_jll v0.4.2+0
[35661453] + Xorg_xkbcomp_jll v1.4.7+0
[33bec58e] + Xorg_xkeyboard_config_jll v2.44.0+0
[c5fb5394] + Xorg_xtrans_jll v1.6.0+0
[8f1865be] + ZeroMQ_jll v4.3.6+0
[3161d3a3] + Zstd_jll v1.5.7+1
[35ca27e7] + eudev_jll v3.2.14+0
[214eeab7] + fzf_jll v0.61.1+0
[a4ae2306] + libaom_jll v3.12.1+0
[0ac62f75] + libass_jll v0.17.4+0
[1183f4f0] + libdecor_jll v0.2.2+0
[2db6ffa8] + libevdev_jll v1.13.4+0
[f638f0a6] + libfdk_aac_jll v2.0.4+0
[36db933b] + libinput_jll v1.28.1+0
[b53b4c65] + libpng_jll v1.6.50+0
[a9144af2] + libsodium_jll v1.0.21+0
[f27f6e37] + libvorbis_jll v1.3.8+0
[009596ad] + mtdev_jll v1.1.7+0
[1270edf5] + x264_jll v10164.0.1+0
[dfaa095f] + x265_jll v4.1.0+0
[d8fb68d0] + xkbcommon_jll v1.9.2+0
[0dad84c5] + ArgTools v1.1.2
[56f22d72] + Artifacts v1.11.0
[2a0f44e3] + Base64 v1.11.0
[ade2ca70] + Dates v1.11.0
[f43a241f] + Downloads v1.6.0
[7b1f6079] + FileWatching v1.11.0
[b77e0a4c] + InteractiveUtils v1.11.0
[b27032c2] + LibCURL v0.6.4
[76f85450] + LibGit2 v1.11.0
[8f399da3] + Libdl v1.11.0
[37e2e46d] + LinearAlgebra v1.11.0
[56ddb016] + Logging v1.11.0
[d6f4376e] + Markdown v1.11.0
[a63ad114] + Mmap v1.11.0
[ca575930] + NetworkOptions v1.2.0
[44cfe95a] + Pkg v1.11.0
[de0858da] + Printf v1.11.0
[3fa0cd96] + REPL v1.11.0
[9a3f8284] + Random v1.11.0
[ea8e919c] + SHA v0.7.0

```

[9e88b42a] + Serialization v1.11.0
[6462fe0b] + Sockets v1.11.0
[2f01184e] + SparseArrays v1.11.0
[f489334b] + StyledStrings v1.11.0
[fa267f1f] + TOML v1.0.3
[a4e569a6] + Tar v1.10.0
[8dfed614] + Test v1.11.0
[cf7118a7] + UUIDs v1.11.0
[4ec0a83e] + Unicode v1.11.0
[e66e0078] + CompilerSupportLibraries_jll v1.1.1+0
[deac9b47] + LibCURL_jll v8.6.0+0
[e37daf67] + LibGit2_jll v1.7.2+0
[29816b5a] + LibSSH2_jll v1.11.0+1
[c8ffd9c3] + MbedTLS_jll v2.28.6+0
[14a3606d] + MozillaCACerts_jll v2023.12.12
[4536629a] + OpenBLAS_jll v0.3.27+1
[05823500] + OpenLibm_jll v0.8.5+0
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.7.0+0
[83775a58] + Zlib_jll v1.2.13+1
[8e850b90] + libblastrampoline_jll v5.11.0+0
[8e850ede] + nghttp2_jll v1.59.0+0
[3f19e933] + p7zip_jll v17.4.0+2

```

Info Packages marked with have new versions available but compatibility constraints

Building Conda → `C:\Users\Donny\.julia\scratchspaces\44cfe95a-1eb2-52ea-b672-e2afdf69b7`

Building IJulia → `C:\Users\Donny\.julia\scratchspaces\44cfe95a-1eb2-52ea-b672-e2afdf69b7`

Precompiling project...

```

878.7 ms LaTeXStrings
863.9 ms TensorCore
875.1 ms Contour
896.6 ms StatsAPI
978.4 ms Measures
980.1 ms Statistics
1065.1 ms Requires
1151.5 ms VersionParsing
1216.6 ms OrderedCollections
843.4 ms StableRNGs
641.3 ms Reexport
1619.4 ms Grisu
772.2 ms PtrArrays
791.1 ms SimpleBufferStream
1080.0 ms Unzip
1184.5 ms DocStringExtensions

```

2079.9 ms	Format
1208.2 ms	URIs
799.1 ms	DataAPI
933.3 ms	DelimitedFiles
946.6 ms	NaNMath
1377.3 ms	TranscodingStreams
808.1 ms	BitFlags
994.4 ms	Scratch
2090.7 ms	IrrationalConstants
1760.0 ms	ConcurrentUtilities
1651.9 ms	LoggingExtras
1542.7 ms	ExceptionUnwrapping
1460.8 ms	Preferences
3902.9 ms	MacroTools
1384.3 ms	Statistics → SparseArraysExt
1034.6 ms	Showoff
2175.5 ms	MbedTLS
885.3 ms	AliasTables
767.0 ms	RelocatableFolders
913.8 ms	Missings
869.5 ms	PrecompileTools
995.2 ms	JLLWrappers
1029.9 ms	CodecZlib
1340.1 ms	LogExpFunctions
3040.5 ms	UnicodeFun
3405.1 ms	FixedPointNumbers
1249.4 ms	Libmount_jll
1209.1 ms	EpollShim_jll
1286.5 ms	Graphite2_jll
1297.6 ms	OpenSSL_jll
1191.7 ms	Bzip2_jll
1204.9 ms	LLVMOpenMP_jll
4242.5 ms	DataStructures
1277.1 ms	libsodium_jll
1307.6 ms	Xorg_libXau_jll
1281.5 ms	LERC_jll
1337.0 ms	Xorg_libICE_jll
1455.9 ms	libfdk_aac_jll
1294.9 ms	LAME_jll
1517.3 ms	libpng_jll
3062.6 ms	RecipesBase
1122.2 ms	fzf_jll
1206.6 ms	JpegTurbo_jll

1095.5 ms	Xorg_libXdmcp_jll
1112.1 ms	mtdev_jll
1269.8 ms	x265_jll
1322.4 ms	XZ_jll
1315.8 ms	Ogg_jll
1341.7 ms	Zstd_jll
1397.3 ms	x264_jll
5315.6 ms	Latexify
1199.1 ms	libaom_jll
1205.8 ms	Expat_jll
1176.3 ms	Opus_jll
1197.2 ms	LZO_jll
1177.3 ms	libevdev_jll
1239.1 ms	Xorg_xtrans_jll
1059.1 ms	eudev_jll
1214.0 ms	Libiconv_jll
1148.1 ms	Libuuid_jll
1172.1 ms	Libffi_jll
1308.5 ms	FriBidi_jll
937.2 ms	SortingAlgorithms
1119.8 ms	Xorg_libSM_jll
1108.8 ms	Xorg_libxcb_jll
1527.2 ms	FreeType2_jll
1399.7 ms	ZeroMQ_jll
1480.2 ms	Pixman_jll
2539.9 ms	ColorTypes
1386.5 ms	libvorbis_jll
1490.0 ms	DBus_jll
1371.2 ms	libinput_jll
2686.9 ms	JLFFzf
1250.7 ms	Wayland_jll
1701.5 ms	Latexify → SparseArraysExt
1240.0 ms	Xorg_xcb_util_jll
3257.3 ms	OpenSSL
1616.4 ms	GettextRuntime_jll
2092.3 ms	Libtiff_jll
1004.1 ms	Xorg_libX11_jll
939.4 ms	Xorg_xcb_util_image_jll
1088.8 ms	Xorg_xcb_util_keysyms_jll
1078.0 ms	Xorg_xcb_util_renderutil_jll
1373.7 ms	ColorTypes → StyledStringsExt
2099.2 ms	Fontconfig_jll
1108.0 ms	Xorg_xcb_util_wm_jll

983.1 ms	Xorg_libXrender_jll
1010.5 ms	Xorg_libXext_jll
1059.1 ms	Xorg_libXfixes_jll
1109.5 ms	Xorg_libxkbfile_jll
3196.7 ms	ColorVectorSpace
1978.6 ms	Glib_jll
1135.7 ms	Xorg_xcb_util_cursor_jll
4661.0 ms	StatsBase
1347.0 ms	Xorg_libXinerama_jll
4130.8 ms	ZMQ
1027.3 ms	Xorg_libXrandr_jll
1015.6 ms	Libglvnd_jll
1240.2 ms	Xorg_libXcursor_jll
1159.4 ms	Xorg_libXi_jll
1159.7 ms	Xorg_xkbcomp_jll
1090.3 ms	Xorg_xkeyboard_config_jll
2533.6 ms	Cairo_jll
1180.9 ms	xkbcommon_jll
7167.5 ms	Colors
15726.5 ms	Parsers
986.3 ms	Vulkan_Loader_jll
2187.8 ms	HarfBuzz_jll
2188.9 ms	JSON
1924.1 ms	libass_jll
1983.4 ms	Pango_jll
2565.4 ms	Qt6Base_jll
1543.3 ms	Conda
921.6 ms	libdecor_jll
1425.9 ms	Qt6ShaderTools_jll
1019.5 ms	GLFW_jll
5421.3 ms	ColorSchemes
3038.6 ms	FFMPEG_jll
2413.2 ms	FFMPEG
2604.5 ms	GR_jll
4033.1 ms	Qt6Declarative_jll
28165.1 ms	Unitful
983.4 ms	Qt6Wayland_jll
1183.5 ms	Unitful → PrintfExt
2398.0 ms	Unitfullatexify
24870.1 ms	HTTP
13936.7 ms	PlotUtils
16502.9 ms	IJulia
4493.2 ms	PlotThemes

```
6605.5 ms    GR
5990.1 ms    RecipesPipeline
67120.3 ms   Plots
5246.6 ms    Plots → UnitfulExt
5932.3 ms    Plots → IJuliaExt
150 dependencies successfully precompiled in 125 seconds. 35 already precompiled.
```

This next cell loads packages which are required for the rest of the code evaluation. In this case, we only need to load the `Plots.jl` plotting package, but you will see others over the course of the semester (and can add more if desired; just make sure that you’ve [added the new packages to the environment](#)). Standard Julia practice is to load all of the needed packages at the top of the file.

Warning

Loading packages can take a while, especially the first time! Julia tries to precompile all of the packages you’re using so repeat use is faster, but this can be quite slow at first.

```
using Plots
```

Introduction

Julia

Julia is an up-and-coming language, originally developed for scientific programming. While learning a new programming language always has its hiccups, the good news is that if you’ve programmed in a high-level language such as Python or MATLAB, most Julia concepts should look familiar.

If you have not successfully set up Julia, follow the instructions in [Tools Setup](#) and/or ask for help.

You can use other editors for this course, but our recommendation is [Visual Studio Code](#) with the [Julia extension](#), which will make life a *lot* simpler! You should have set this up by following the [Tools Setup](#) instructions, but if not, do so now and/or ask for help.

Jupyter Notebooks

Jupyter notebooks integrate text and equations in Markdown with Julia (or Python, or R) code. To do this, Jupyter notebooks consist of two types of “cells”: code cells and Markdown (text) cells.

Click once on this section of text. A box will appear around this text (and some areas above/below it) - all of that is within this cell.

Markdown is a text markup framework for formatting language that makes things look pretty when viewed across different platforms: web browsers, notebooks, and so forth. Text written in Markdown can also include hyperlinks, LaTeX equations, section headers, and images, among other features (most of [the course website](#) and the lecture notes were all written in Markdown!). [Here is a basic Markdown cheat sheet](#).

What you are looking at right now is the formatted text after the Markdown is processed. To see the raw Markdown, do one of:

- press **Enter** while that cell is selected, or
 - double-click on that cell.
-

A couple of the features you will see in this Markdown cell:

- The `---` command creates a horizontal line. This is also nice for separating sections.
- Backticks (``...``) can be used to format and highlight code, keystrokes, etc.
- The `#` sign is used to create a new section header; two `#` signs (`##`) is used to create a new subsection header; `###` creates a subsubsection, and so on.
- You can create a bulleted list by using the asterisk `*` or a dash `-` and a space.
- You can create regular text by just typing as usual.
- You can create **bold-faced text** by wrapping it with two asterisks on both sides.
- You can create *italicized text* by wrapping it with a single asterisk on both sides.
- To create a new paragraph, you must include a blank line between the old and new paragraphs.

At this point you might be wondering how to turn this cell back into the fully formatted Markdown text instead of the raw Markdown you’re probably still looking at. You have a couple of options, depending on your platform, but the most consistent is to type **Shift + Enter** to **execute** the cell (this is also how to run code, but more on that later).

Additionally, you will frequently need to create new cells in your Jupyter notebooks. How you do this will depend on how you interact with the notebook, but try to figure this out now.

One tip is to think carefully about what bits of code should be in the same cell, as you typically only see output from the last command in a cell. For example, compare the following:

```
x = 5  
sin(x)
```

-0.9589242746631385

with

```
x = 5
```

5

```
sin(x)
```

-0.9589242746631385

In Julia, you can also suppress the output of a command with a semi-colon:

```
sin(x);
```

which can help if you want to split some code out for clarity or to insert some text prior to it, but don't want to clutter the notebook with its output.

For code cells, to execute the commands within the cell, we also press **Shift+Enter**.

Finally, **make sure that you evaluate all of the code cells in order before submitting**. One bad outcome with notebooks occurs when cells are evaluated out of order, so fixed bugs and edits in previous cells do not get a chance to propagate down. You can do this with the **Run All** command in whichever interface you're using to edit your notebook.

Julia Basics

There are many tutorials and references for Julia, including a [basics overview on the class website](#). Please feel free to reference these as you work through any part of the course.

Formatting Math

It will often be helpful to include nicely-formatted mathematics in a notebook. Markdown accomodates this using LaTeX syntax. A LaTeX cheatsheet is available on the class website, and many other guides exist online.

Below is an example of a formatted equation:

$$x = 5.$$

Looking For Help

There is no shame in using Google, or other resources, for help when programming. There are many, many times when you can't quite get the syntax to work, can't quite figure out the right package or command to use, or are feeling too lazy or overwhelmed (I'm not judging either of those!) to dig through the documentation. Some good resources include:

- [Stack Overflow](#) is a treasure trove of answers;
- The [official Julia forum](#) and the [Julia Subreddit](#) are also very useful.

You are also highly encouraged to post on [Ed Discussion](#), though getting a response might be less immediate. Just be mindful that to get good answers, [you have to help people help you](#), and **make sure to give credit to any resources that were helpful!**

Exercises (3 points)

Use your understanding of Julia syntax and the GitHub workflow to complete the following (hopefully short) exercises. Convert your completed lab assignment to a PDF and submit it to the Gradescope Assignment "Lab 1".

Remember to:

- Include a (succinct but clear) writeup of the core idea underlying your code, through some combination of equations, text, and algorithms. As you are not required to submit your code, we will not be looking at it in detail, and instead will rely on those writeups to assess whether your approach is correct.
- If using the notebook, evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Computing a Dot Product

Given two numeric arrays `x` and `y`, write a function to compute their dot product if they have equal length, and return an error if not (this is useful for debugging!). Use the following code as a starting point. **Do not use a built-in function.**

```
function dot_product(x, y)
    if x==y
        return dot_product
    else
```

```

        throw(DimensionMismatch("length of x not equal to length of y"))
    end
end

```

`dot_product` (generic function with 1 method)

Here are some tests to make sure your code works as intended. Tests like these are useful to make sure everything works as intended. One reason to split your code up into functions is that it makes it straightforward to write tests to make sure each piece of your code works, which makes it easier to identify where errors are occurring.

```
dot_product([1 2 3], [4 5 6])
```

```

DimensionMismatch: DimensionMismatch: length of x not equal to length of y
DimensionMismatch: length of x not equal to length of y

```

Stacktrace:

```

[1] dot_product(x::Matrix{Int64}, y::Matrix{Int64})

   @ Main c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X15sZm
[2] top-level scope

   @ c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X20sZmlsZQ

```

If you know the value you should get, you can write a more formal test using the `@assert` macro, which is a good way to “automate” checking (since you get an obvious error if the code doesn’t work as desired):

```
@assert dot_product([1 2 3], [4 5 6]) == 32
```

```

DimensionMismatch: DimensionMismatch: length of x not equal to length of y
DimensionMismatch: length of x not equal to length of y

```

Stacktrace:

```
[1] dot_product(x::Matrix{Int64}, y::Matrix{Int64})
```

```
@ Main c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X15sZ
```

```
[2] top-level scope
```

```
@ c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X22sZmlsZQ
```

Let's also make sure we get an error when the dimensions of the two vectors don't match:

```
dot_product([1 2 3], [4 5])
```

```
DimensionMismatch: DimensionMismatch: length of x not equal to length of y  
DimensionMismatch: length of x not equal to length of y
```

Stacktrace:

```
[1] dot_product(x::Matrix{Int64}, y::Matrix{Int64})
```

```
@ Main c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X15sZ
```

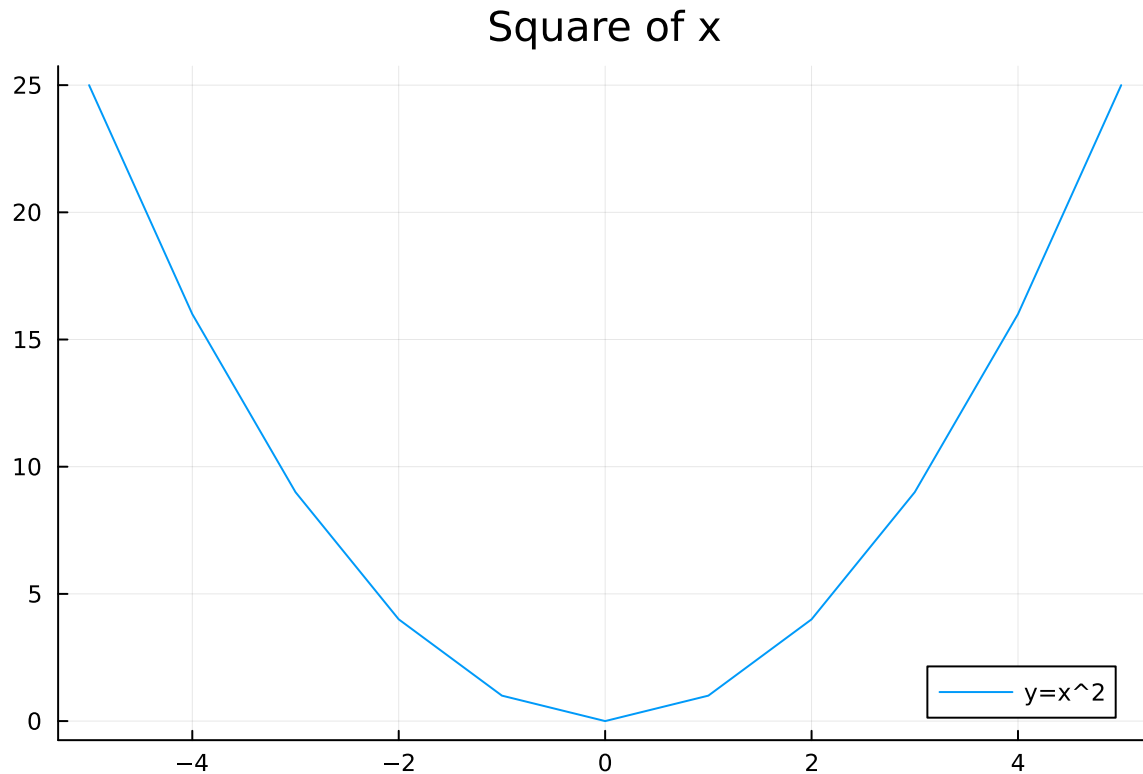
```
[2] top-level scope
```

```
@ c:\Users\Donny\lab1-dc2266\jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X24sZmlsZQ
```

Making a Plot

Write a function to compute the square of an integer x . Evaluate this function for integers between $x = -5$ and $x = 5$ and make a plot of the squared values (you can find a quick guide to making various types of plots [here](#)). Make sure to label your axes.

```
# insert your code here  
x_integer = -5:5  
y_integer = x_integer.^2  
p = plot(x_integer, y_integer, label="y=x^2", title="Square of x")  
p;
```



Commit and Push Your Changes to GitHub

After completing the previous two exercises, commit your solution file (notebook or otherwise) and push to GitHub. Use an informative commit message which makes it clear what changes you've made. The specific workflow for this will vary depending on how you're writing up your solutions; please search for specifics and ask for help as needed!

Useful Commit Sizes

Ideally, you'd commit whenever you make a “substantial” enough change that you want to lock in, such as writing the core code for a problem or completing a problem, if you're preparing code to be used elsewhere (by yourself or others), or if you want to ask for help. `git` lets you revert changes back to a previous commit, so it's easy to undo changes or updates which broke something that was previously working, so changing too many things at once can make it hard to keep track of what worked when.

But in this case, go ahead and just commit after finishing the problems.

Push the repository with these commits to GitHub and take a screenshot of the repository page (<https://github.com/BEE4750-FA25/<username>/lab01>) which shows the updated repository. Include that screenshot in your submission as the solution to this problem.

Submitting PDF

Important

These submission instructions will not be repeated on future assignments!

Export your writeup as a PDF (this doesn't have to be a notebook if you're struggling to export: you can do your writeup in *e.g.* Microsoft Word) and submit it to the "Lab 1" assignment on Gradescope. **Make sure that you tag pages corresponding to relevant problems to avoid a 10% penalty.**

Printing Code to PDF

You are not required to submit your code when submitting assignments. However, when printing a notebook to PDF (whether you do this locally or via the GitHub Action), long lines will run off the edge of code cells, which may result in comments or code being hidden. If you see this, go back to the notebook and break up long lines into shorter ones (for example, see the comment in the above code cell) to ensure key parts of your results aren't missing.